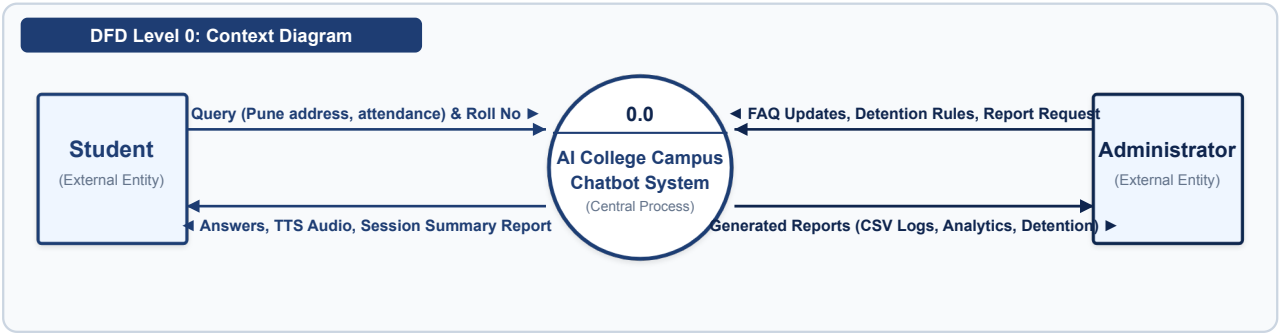
GHRaisoni International Skill Tech University, Pune

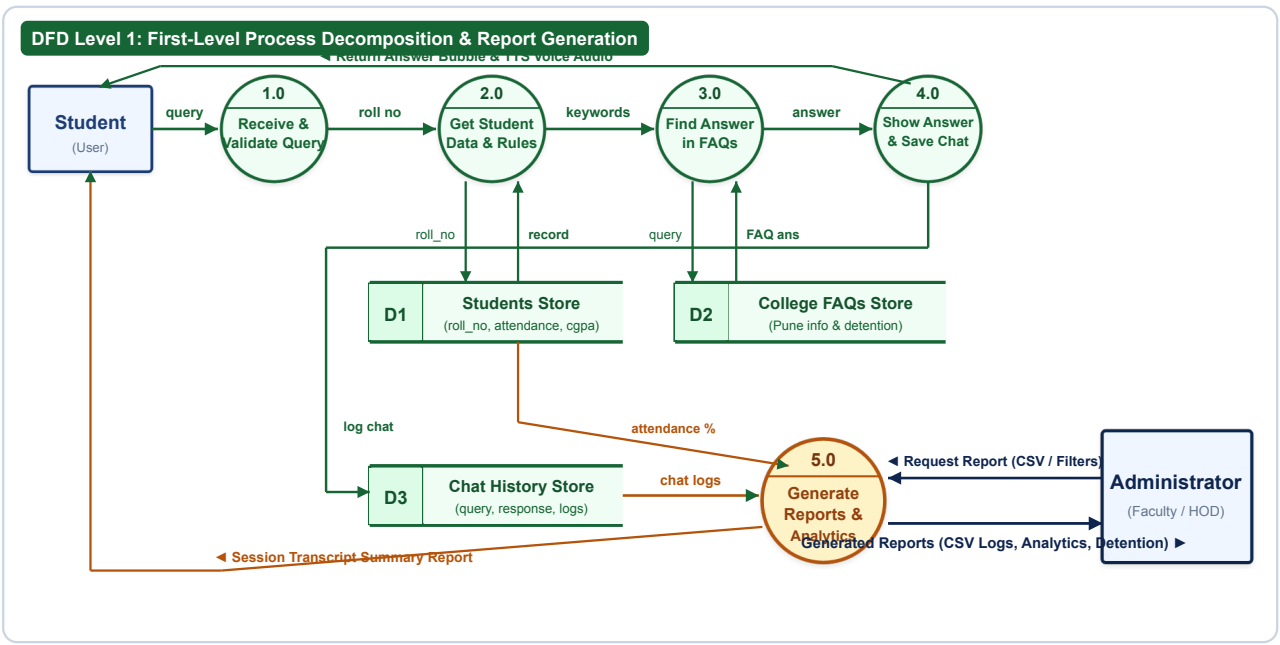
AI CHATBOT FOR COLLEGE

FACULTY REVISED SYSTEM ARCHITECTURE & DFD SPECIFICATION

✓ Context-Level DFD (Level 0) Included ✓ Standard Open-Ended Data Stores ✓ Circle Process Bubbles (DeMarco/Gane-Sarson)

6. Data Flow Diagram (DFD Level 0 Context & Level 1 Process Decomposition with Report Generation)





Standard DFD Notation & Symbol Conventions (Gane-Sarson / DeMarco)

Symbol Shape	DFD Element	Standard Academic Definition	Implementation in Our Project
▭	External Entity	Origin/destination of data outside boundary (Square / Rectangle)	Student (User) & Administrator (Faculty/HOD)
◯	Process (Bubble)	Transforms incoming data into outgoing data (Circle with ID)	1.0 to 4.0 Query Handlers & 5.0 Generate Reports
D ═	Data Store (Open)	Repository of data at rest (Open-ended parallel lines; NEVER closed)	D1 Students, D2 College FAQs, D3 Chat History

Fig 6: Data Flow Diagram (DFD) — Top: Level 0 Context Diagram showing system boundary with Student & Administrator; Middle: Level 1 Process Decomposition (1.0 to 4.0 Query Handlers & 5.0 Generate Reports & Analytics) with standard open-ended parallel line data stores (D1 Students, D2 FAQs, D3 Chat History); Bottom: Standard DFD notation and symbols reference table.